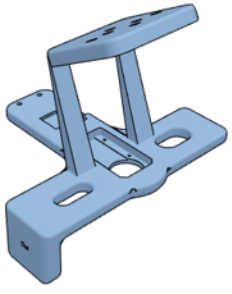
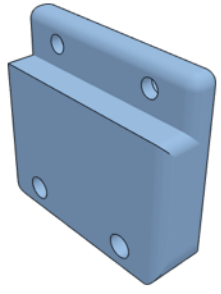
Parts

Print these (hardware/STEP or STL, right/ for the right hand, left/ for the left).



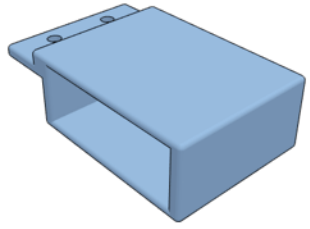
main_support

1×



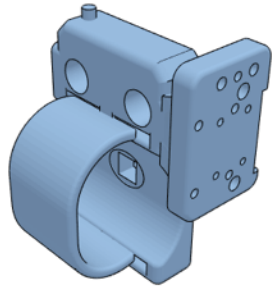
main_support_cover_plate

1×



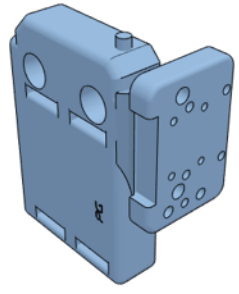
hand_support_base

1×



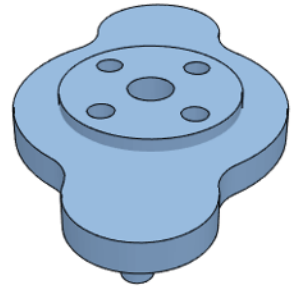
index_link

1×



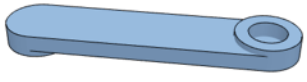
thumb_link

1×



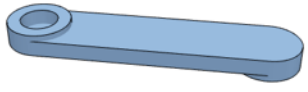
crank_mechanism_plate

1×



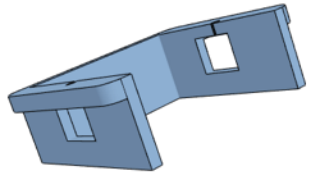
connecting_link_1

1×



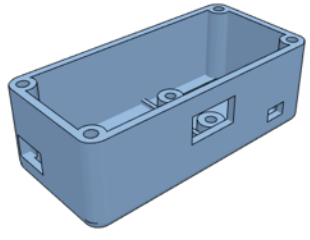
connecting_link_2

1×



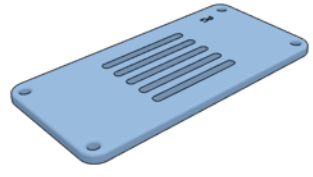
d405_hood

1×



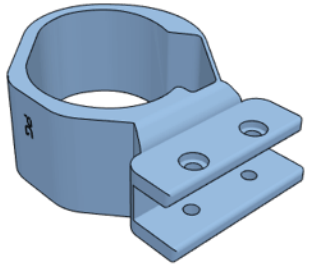
electronics_box

1×



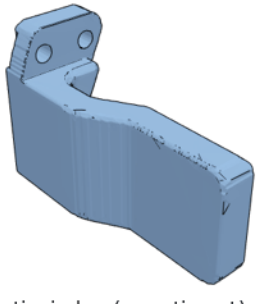
electronics_lid

1×



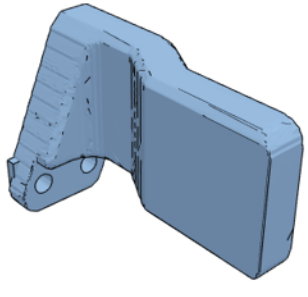
controller_support

1×



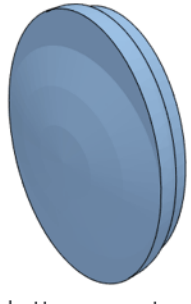
tip_index (your tip set)

1×



tip_thumb (your tip set)

1×



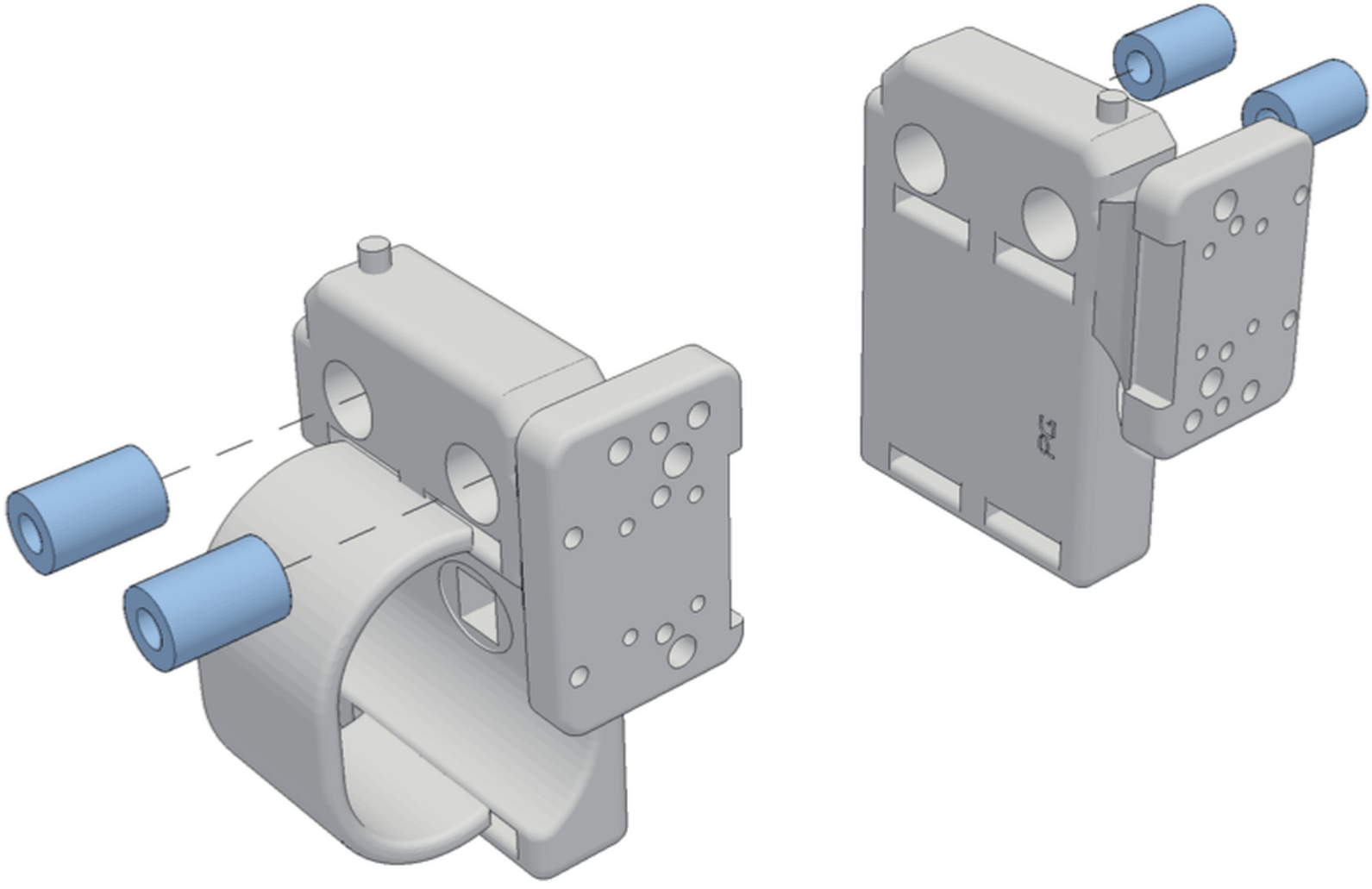
button_cover_tpu

1×

1

Linear bearings

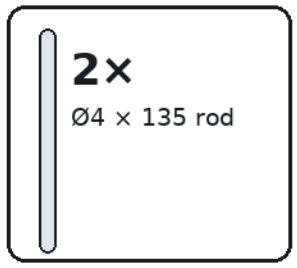
4×
LM4UU 4 × 8 × 12

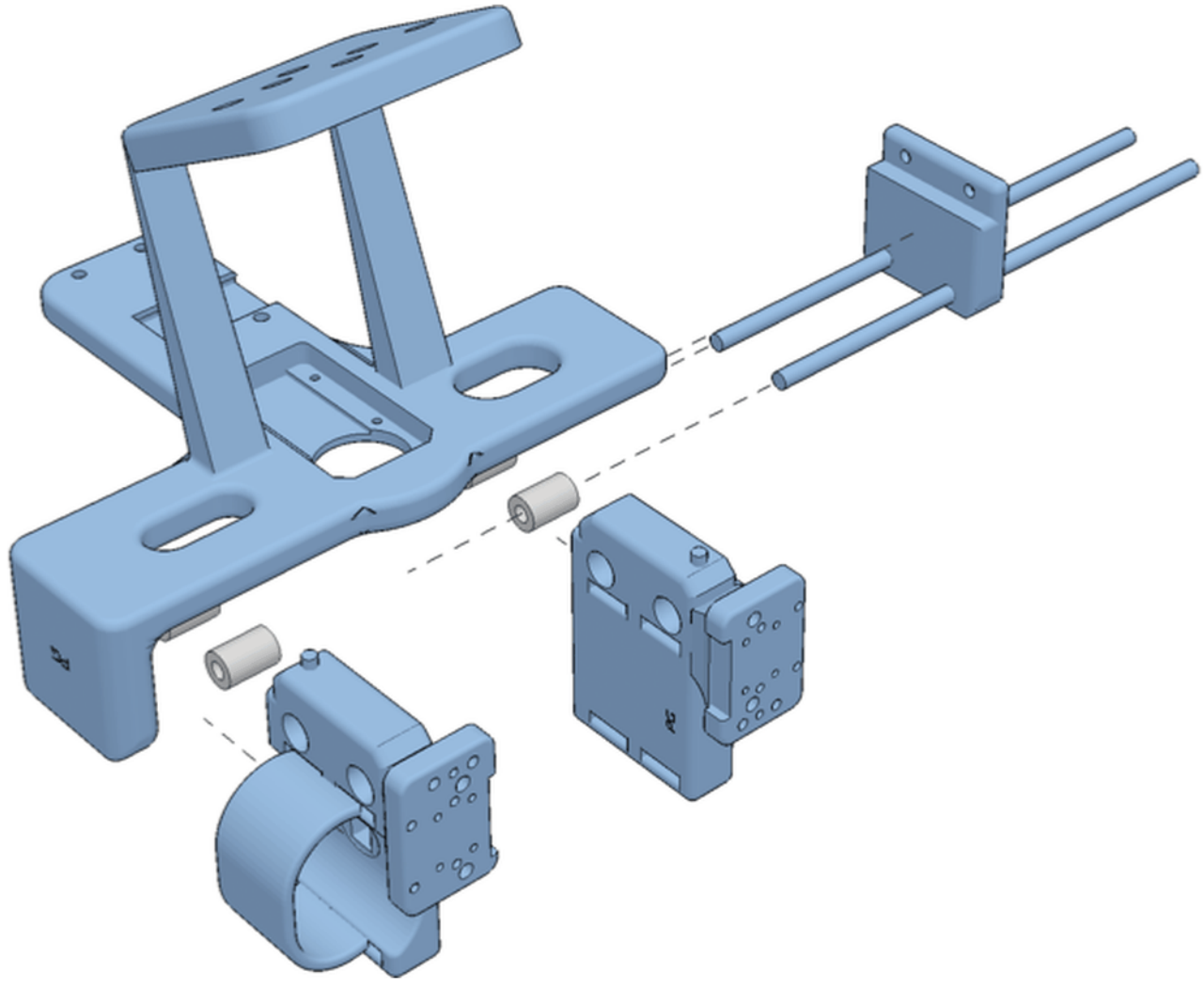


Press two LM4UU linear bearings into each finger link (index and thumb).

2

Rods and links



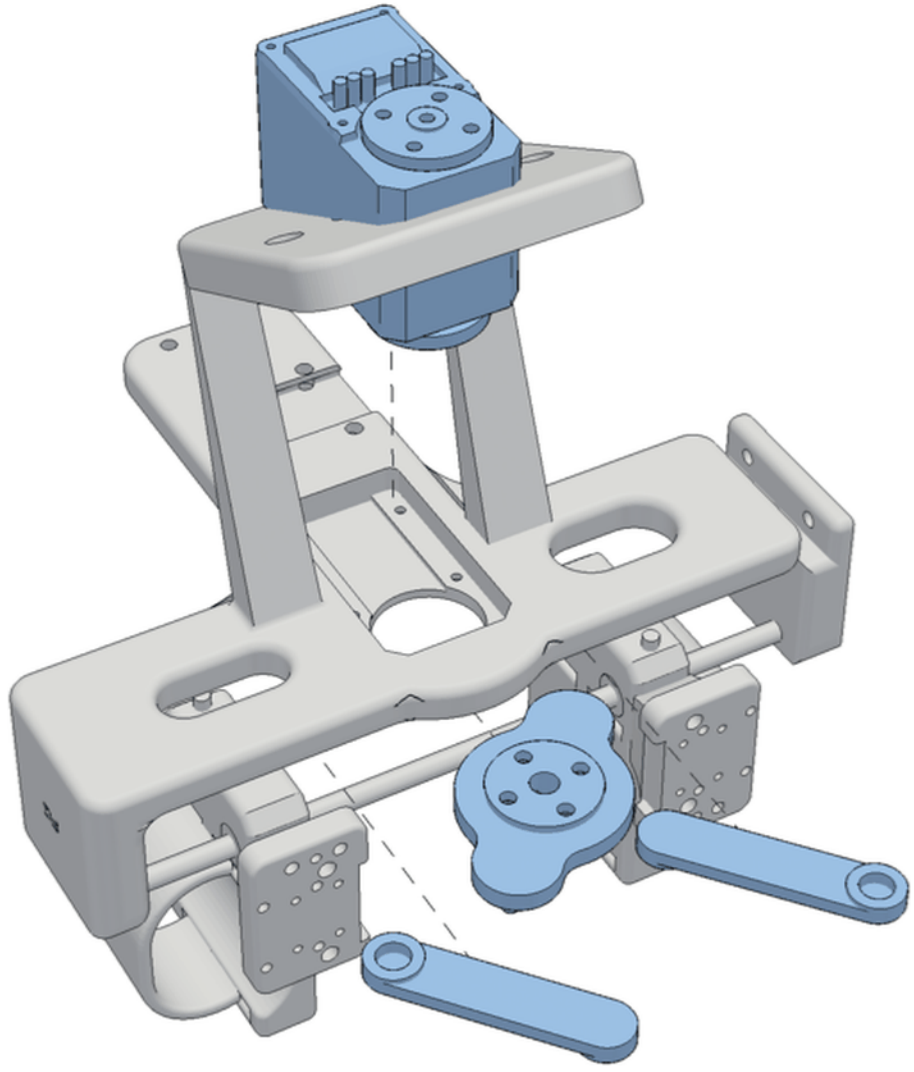


Hold both links in the main support and push the two Ø4 rods through the support and the links' bearings. Close the rod ends with the cover plate.

3

Encoder and linkage

1×	4×	4×	4×	2×
STS3215 servo	MR63ZZ 3 × 6 × 2.5	M3 × 12	M3 × 15	M3 × 20

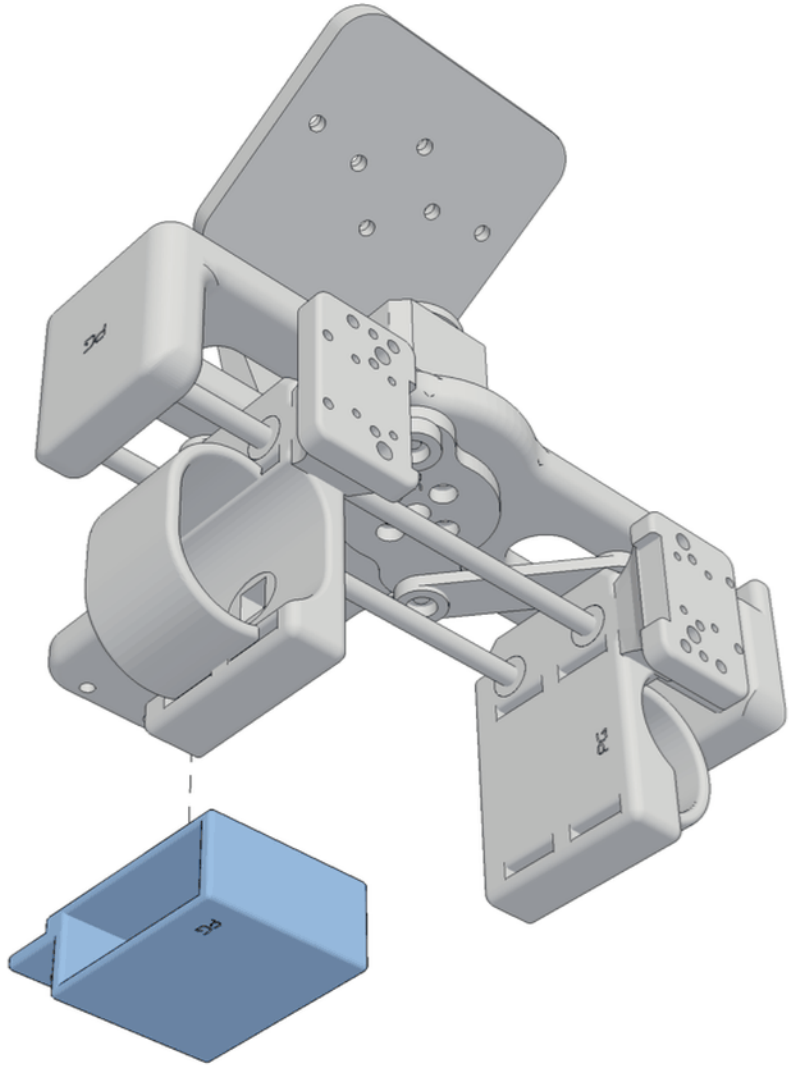


Seat the STS3215 (encoder, torque off) in the support. Fit the crank on its horn and join it to both links with the connecting links on MR63ZZ bearings. Screws: HandUMI's mechanism set.

4

Hand support

4×
hook-and-loop strap

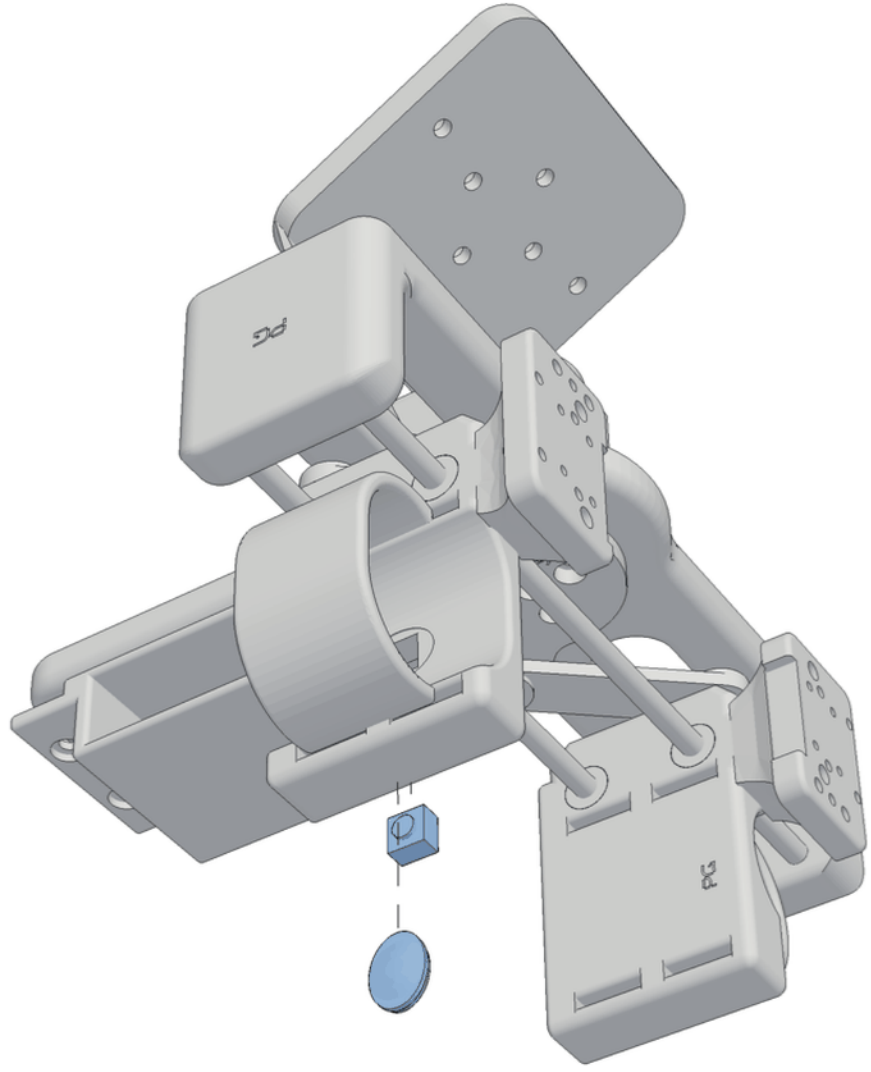


Screw the hand support under the main support and thread the hook-and-loop straps through it.

5

Record button

1×
6 × 6 tactile switch + wire



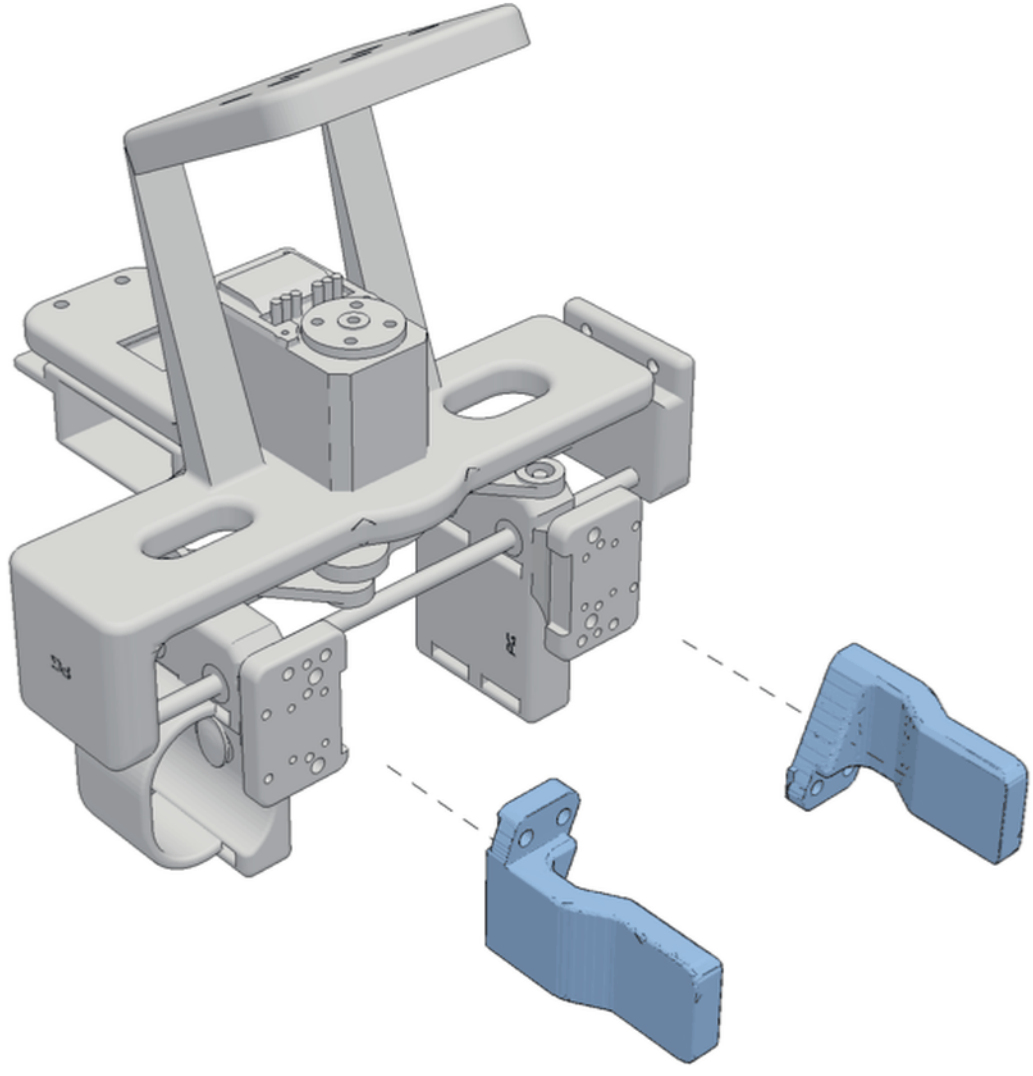
Lay the switch's wire in the index link's channel, press the switch in and push the TPU cover over it.

6

Tips

4× M3 × 10

2× Ø2 × 6 dowel (optional)

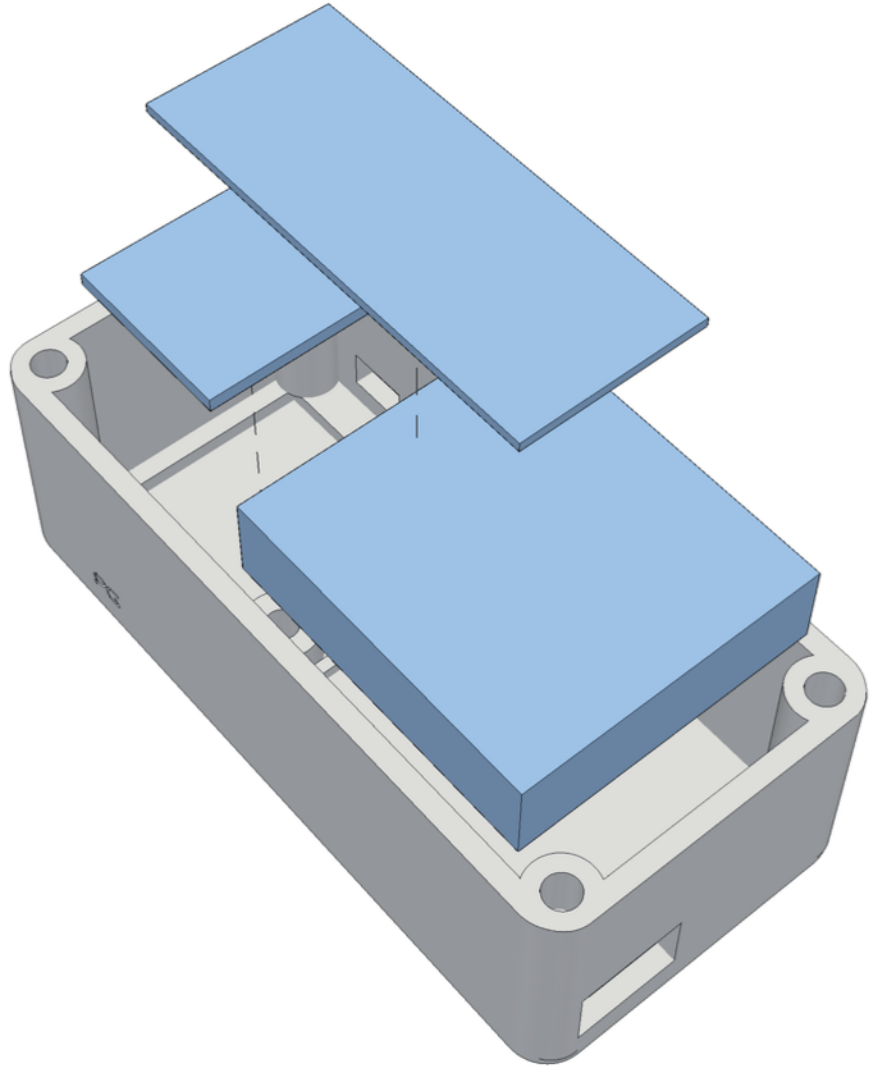


Screw your robot's fingers onto the two links with the robot's own screws (here: Robotiq Hand-E, 2 × M3 × 10 each, into the Hand-E holes at opposite ends of the two flanges). Piper: 4 × M2 × 10; YAM: 2 × M4 × 10; OpenArm: 4 × M2.5 × 8. Real Robotiq fingers go on over Hand-E-Rack-Shim.

7

Electronics

4×
M2 screws (Pico)

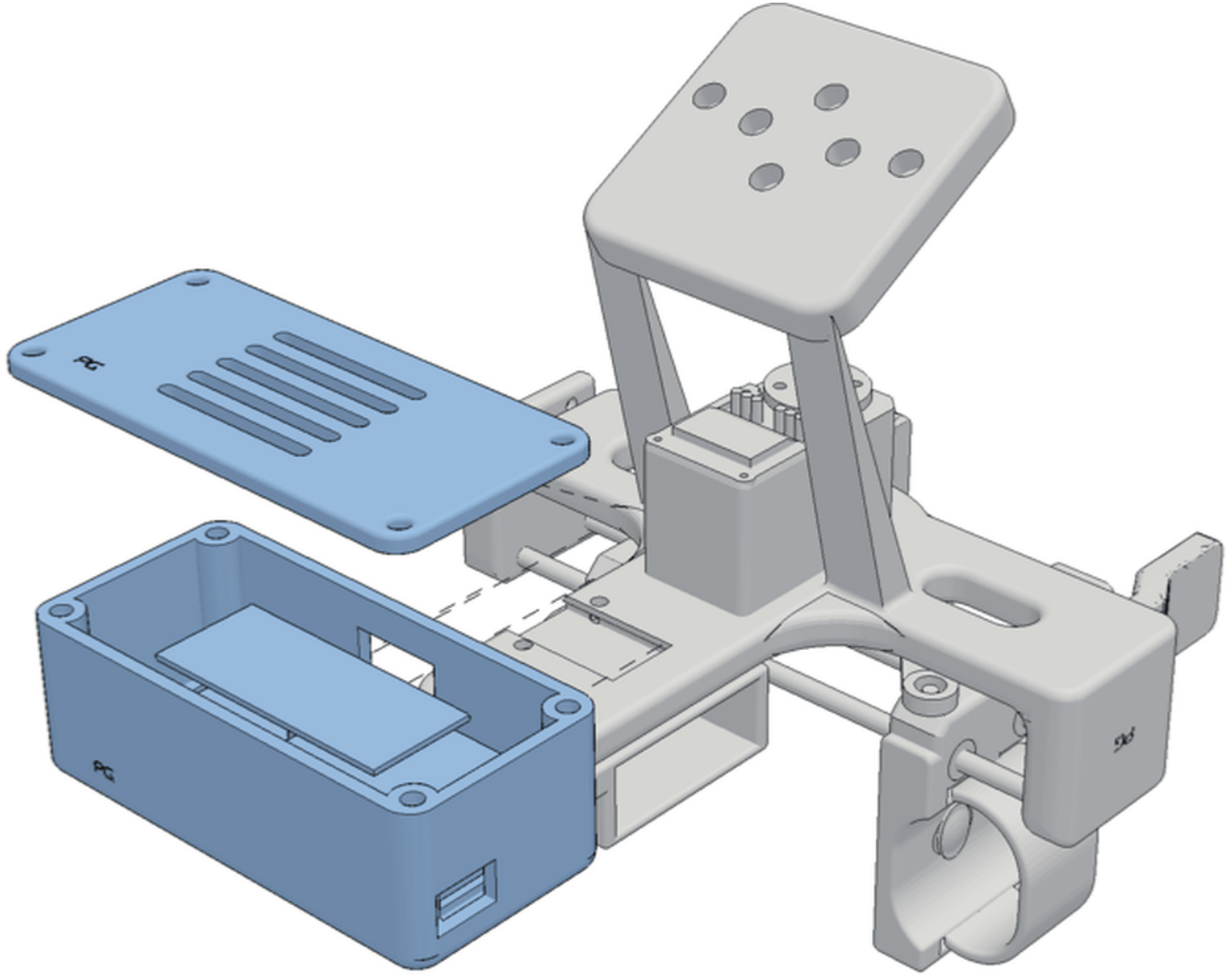


Fit the Pico 2, the IMU and the USB 3 hub in the electronics box and wire them as in docs/electronics.md.

8

Electronics box

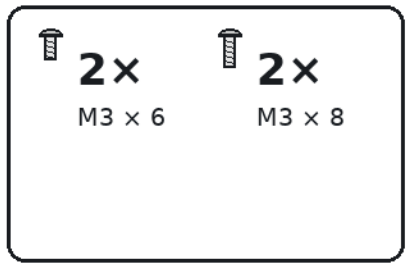
4×	4×	8×
M3 × 8	M3 × 6 (lid)	M3 heat-set insert

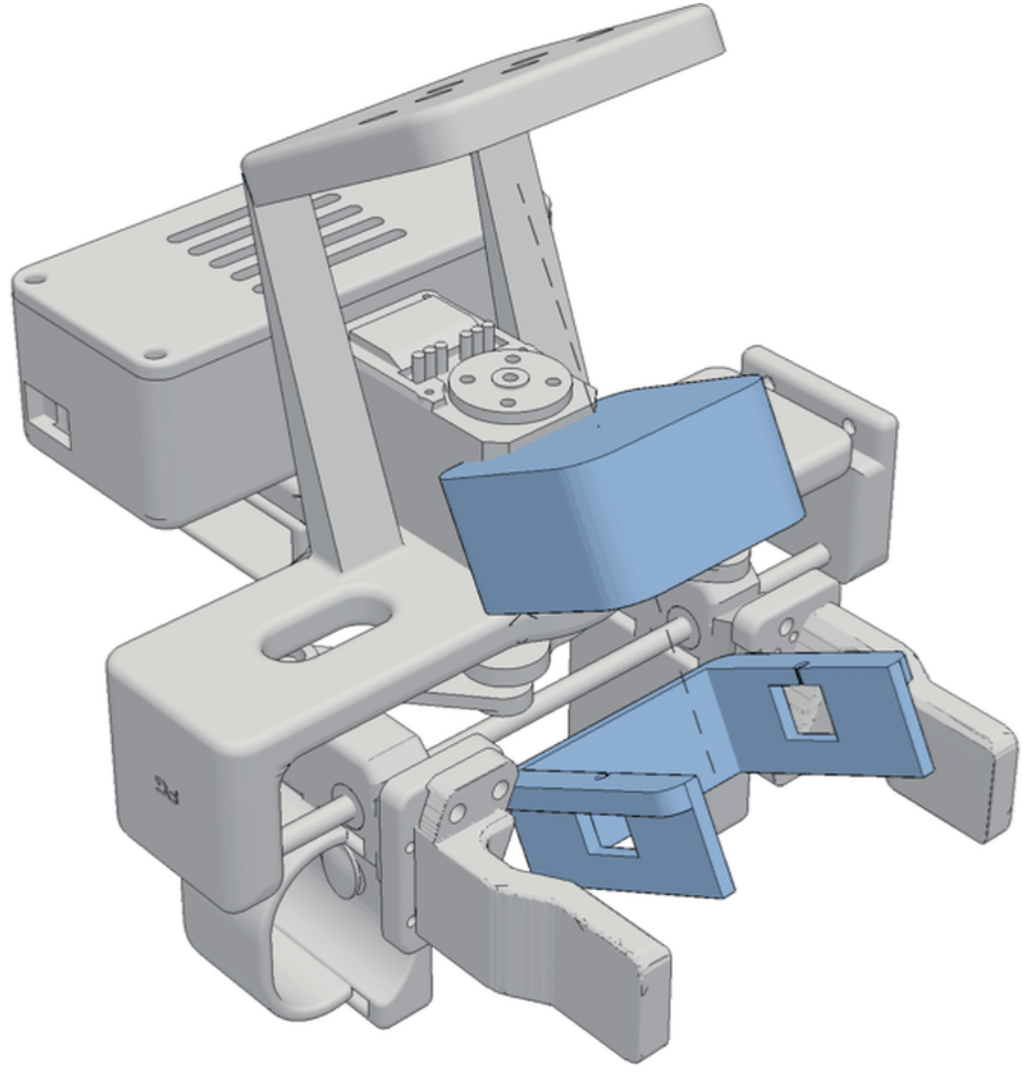


Screw the box to the support's arm (4 × M3 × 8), then close the lid (4 × M3 × 6 into inserts).

9

Camera

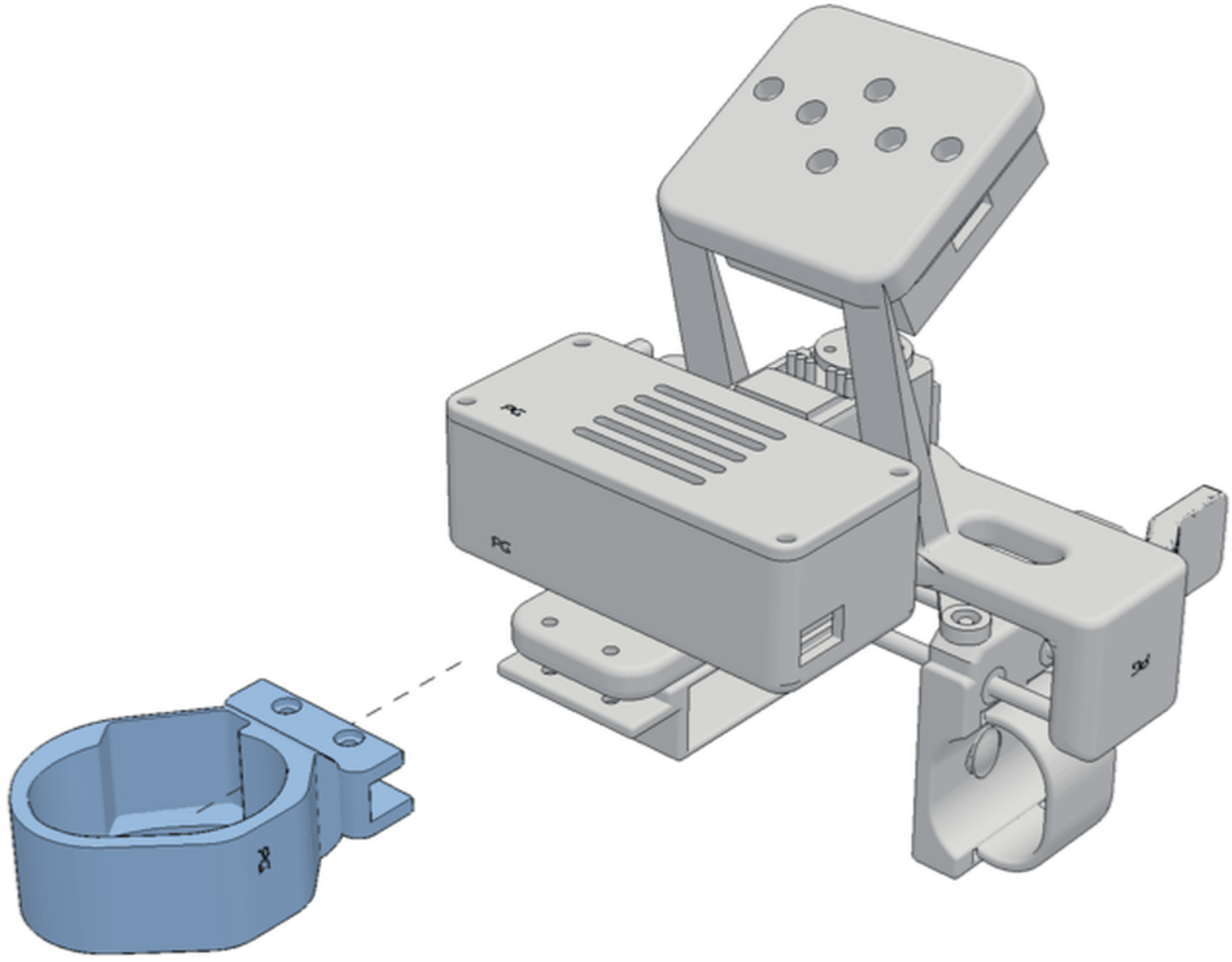




Set the D405's back on the tab at the top of the arch and screw it on from behind (2 × M3 × 6 into its rear holes; no longer, the threads are 4 mm deep). Slide the hood over it and fix it from behind through the tab's outer holes (2 × M3 × 8). The robot's hande_camera_plate takes the same camera, hood and screws.

10

Controller support



Bolt the controller support to the back of the main support and push the Quest controller into its ring.